

Aula 2 - Laboratório de Física 1

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1X12

Resumo

Média:

$$\bar{x} = \sum_{i=1}^N \frac{x_i}{N}$$

Desvio Padrão:

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\bar{x} - x_i)^2}$$

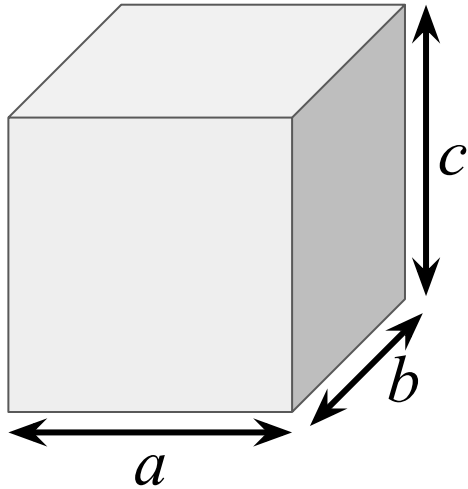
Erro Estatístico:

$$\sigma_{\bar{x}} = \Delta x_{estat} = \frac{\sigma}{\sqrt{N}}$$

Erro Total:

$$\Delta x_{total} = \sqrt{(\Delta x_{estat})^2 + (\Delta x_{instr})^2}$$

Propagação de erro



$$a = (5,62 \pm 0,05) \text{ cm}$$

$$b = (4,06 \pm 0,05) \text{ cm}$$

$$c = (6,89 \pm 0,05) \text{ cm}$$

$$V = (5,62 \times 4,06 \times 6,89 \pm ?) \text{ cm}^3$$

$$V = (157,21 \pm ?) \text{ cm}^3$$

Propagação de erro

Considere uma grandeza u que esteja relacionado com outras grandezas $x_1, x_2, x_3, \dots, x_n$:

$$u = f(x_1, x_2, x_3, \dots, x_n)$$

Cada grandeza x_i apresenta sua incerteza σ_i correspondente, isto é, cada grandeza é mostrada na forma $x_i \pm \sigma_i$.

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A incerteza do parâmetro u é dada por

$$\sigma_u^2 = \left(\frac{\partial u}{\partial x_1} \right)^2 \sigma_{x_1}^2 + \left(\frac{\partial u}{\partial x_2} \right)^2 \sigma_{x_2}^2 + \dots + \left(\frac{\partial u}{\partial x_n} \right)^2 \sigma_{x_n}^2$$

Propagação de erro

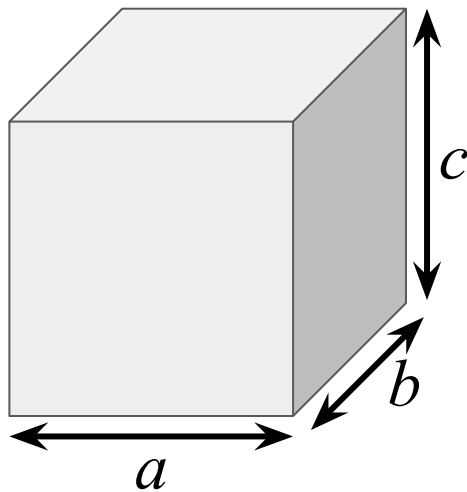
Exemplo: volume de um paralelepípedo

$$V = V(a, b, c) = abc$$

$$\begin{aligned}\sigma_V^2 &= \left(\frac{\partial V}{\partial a}\right)^2 \sigma_a^2 + \left(\frac{\partial V}{\partial b}\right)^2 \sigma_b^2 + \left(\frac{\partial V}{\partial c}\right)^2 \sigma_c^2 \\ &= (bc)^2 \sigma_a^2 + (ac)^2 \sigma_b^2 + (ab)^2 \sigma_c^2\end{aligned}$$

$$\left(\frac{\sigma_V}{V}\right)^2 = \left(\frac{\sigma_a}{a}\right)^2 + \left(\frac{\sigma_b}{b}\right)^2 + \left(\frac{\sigma_c}{c}\right)^2$$

Propagação de erro



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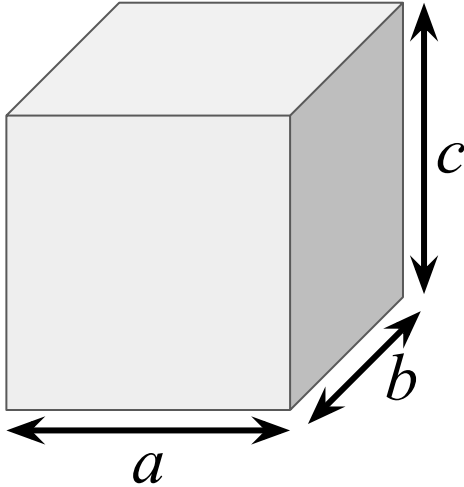
$$V = (5,62 \times 4,06 \times 6,89 \pm ?) \text{ cm}^3$$

$$V = (157,21 \pm ?) \text{ cm}^3$$

$$\begin{aligned} \sigma_V^2 &= (4,06 \times 6,89)^2 (0,05)^2 + \\ &\quad (5,62 \times 6,89)^2 (0,05)^2 + \\ &\quad (5,62 \times 4,06)^2 (0,05)^2 = 7,00 \end{aligned}$$

$$\sigma_V = 2.64 \sim 3$$

Propagação de erro



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Linearização

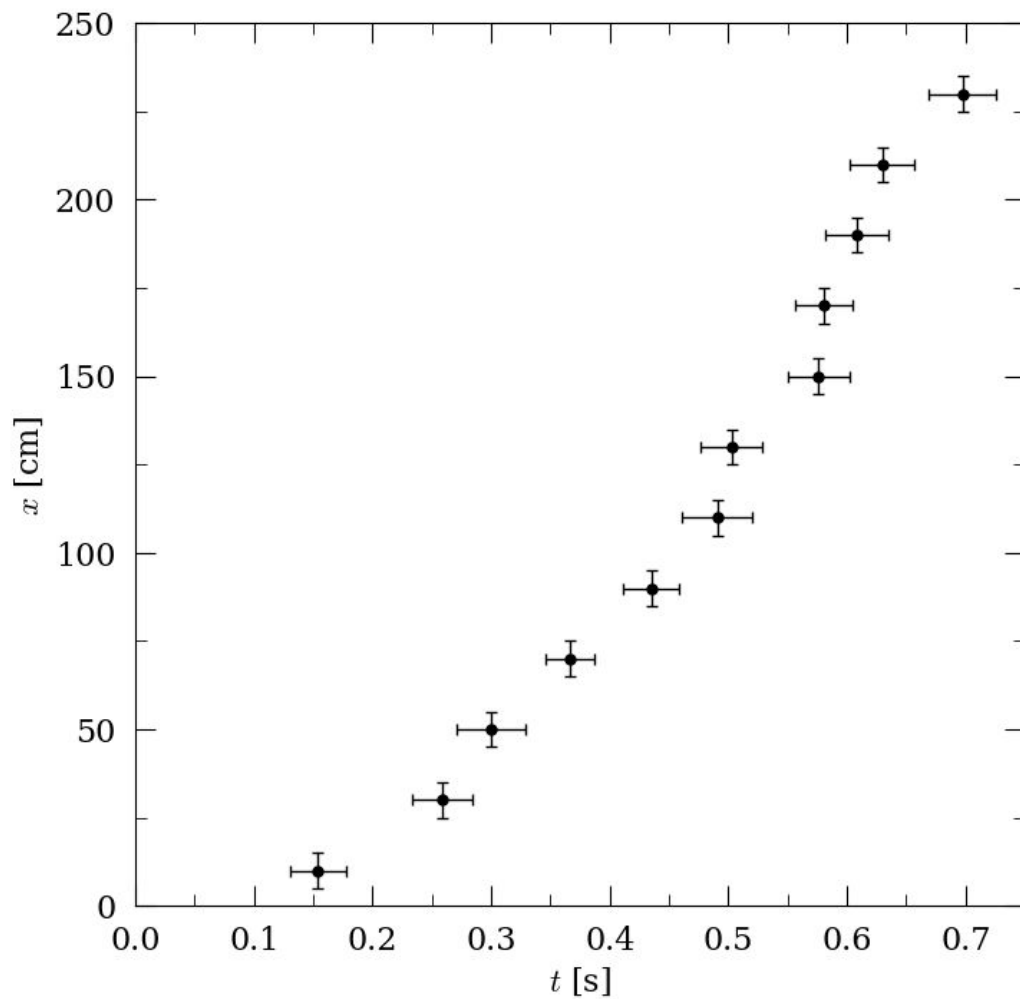
Frequentemente observamos em ciência que:

$$y = ax^b$$

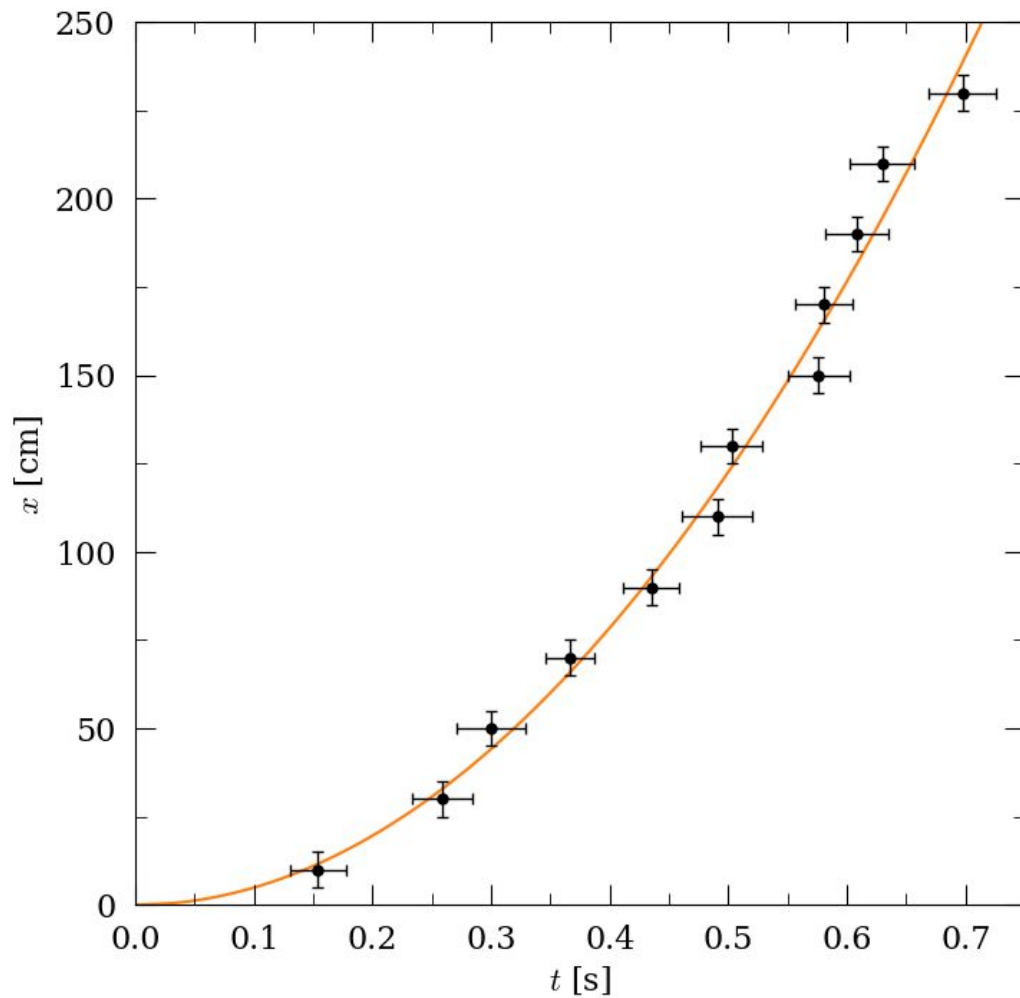
Por exemplo a equação da queda livre

$$y = \frac{1}{2}gt^2$$

$t + \Delta t$ [s]	$y + \Delta y$ [cm]
0.15 ± 0.02	10 ± 5
0.26 ± 0.03	30 ± 5
0.30 ± 0.03	50 ± 5
0.37 ± 0.02	70 ± 5
0.44 ± 0.02	90 ± 5
0.49 ± 0.03	110 ± 5
0.50 ± 0.03	130 ± 5
0.58 ± 0.03	150 ± 5
0.58 ± 0.02	170 ± 5
0.61 ± 0.03	190 ± 5
0.63 ± 0.03	210 ± 5
0.70 ± 0.03	230 ± 5



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Linearização

Como obter constante a e expoente n ?

$$y = ax^b$$

Resposta: linearização

$$\begin{array}{l} \ln y = \ln(ax^b) \\ \underbrace{\ln y}_Y = \underbrace{\ln a}_A + b \underbrace{\ln x}_X \end{array} \implies Y = A + bX$$

Linearização

$$y = ax^b$$

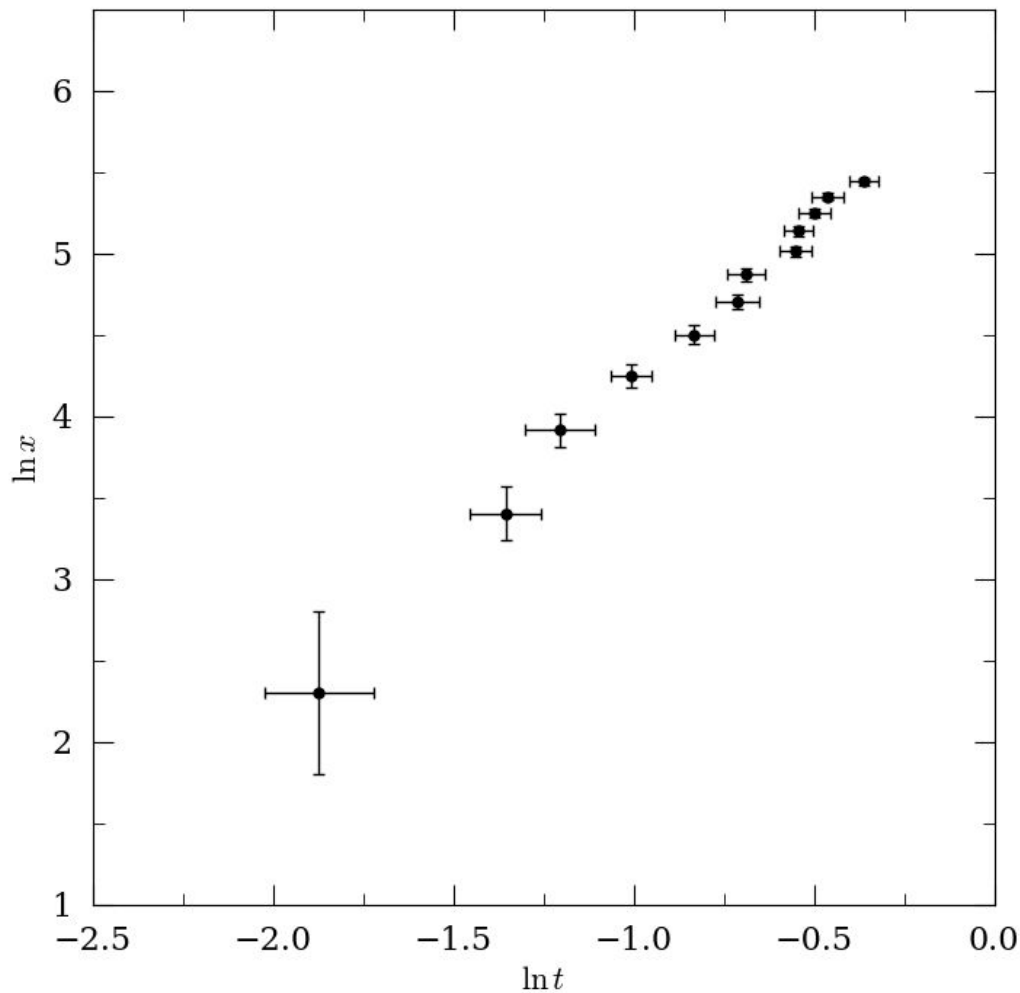
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Erro da linearização

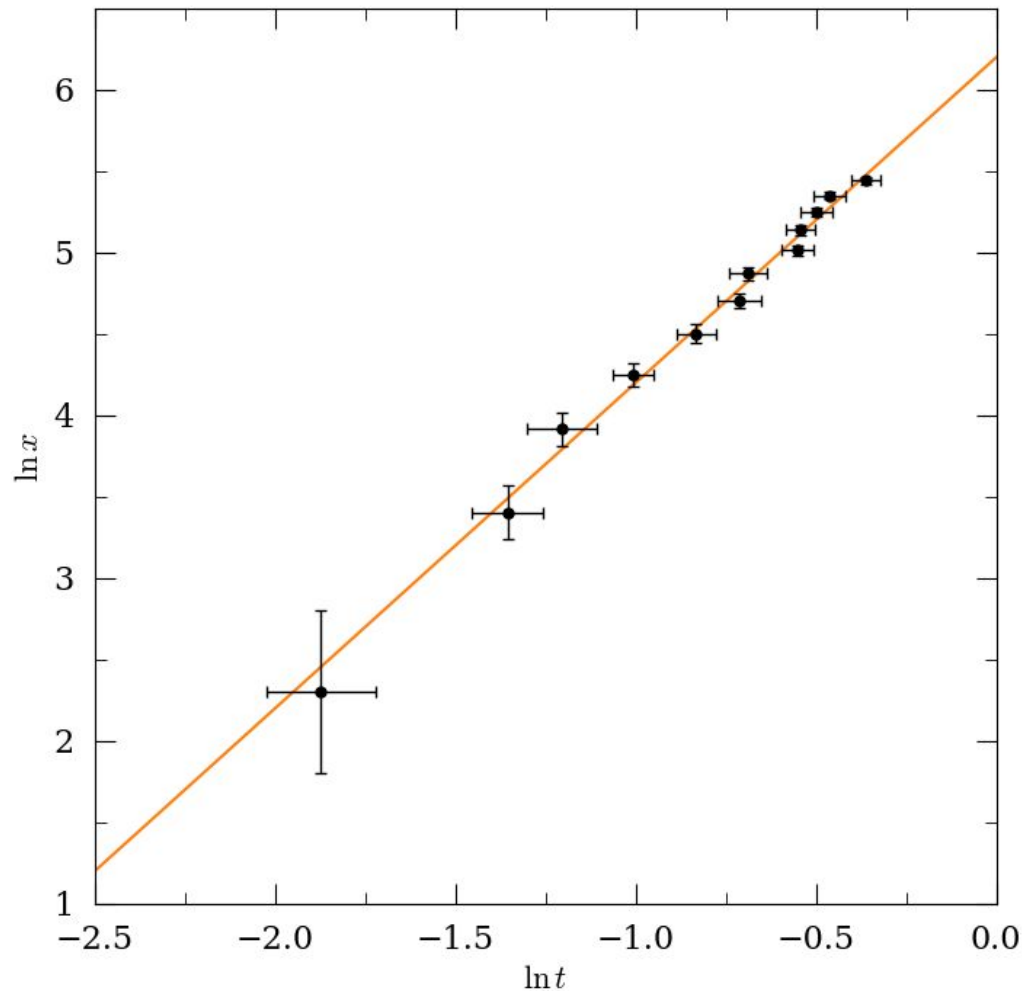
$$\sigma_{\ln x} = \sqrt{\left(\frac{\partial \ln x}{\partial x}\right)^2 (\sigma_x)^2} = \left|\frac{\sigma_x}{x}\right|$$

$$\sigma_{\ln y} = \sqrt{\left(\frac{\partial \ln y}{\partial y}\right)^2 (\sigma_y)^2} = \left|\frac{\sigma_y}{y}\right|$$

$\ln t + \Delta t$ [s]	$\ln y + \Delta y$ [cm]
-1.9 ± 0.2	2.3 ± 0.5
-1.4 ± 0.1	3.4 ± 0.2
-1.2 ± 0.1	3.9 ± 0.1
-1.00 ± 0.06	4.25 ± 0.07
-0.83 ± 0.05	4.50 ± 0.06
-0.71 ± 0.06	4.70 ± 0.05
-0.69 ± 0.05	4.87 ± 0.04
-0.55 ± 0.04	5.01 ± 0.03
-0.54 ± 0.04	5.14 ± 0.03
-0.50 ± 0.04	5.25 ± 0.03
-0.46 ± 0.04	5.35 ± 0.02
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Método de mínimos quadrados para Regressão Linear

$$a = \frac{\left(\sum_{i=1}^n w_i\right) \left(\sum_{i=1}^n w_i y_i x_i\right) - \left(\sum_{i=1}^n w_i y_i\right) \left(\sum_{i=1}^n w_i x_i\right)}{\Delta} \quad w_i = \frac{1}{\sigma_i^2}$$

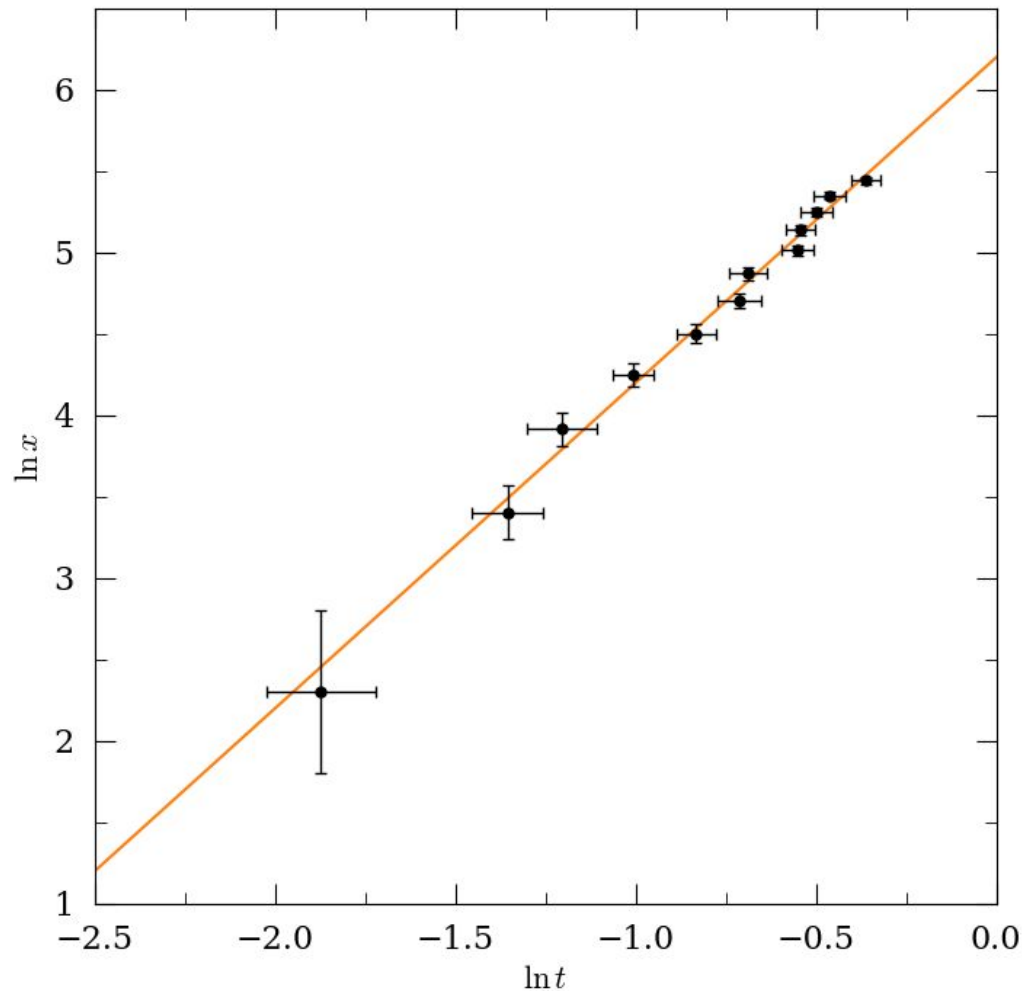
$$b = \frac{\left(\sum_{i=1}^n w_i y_i\right) \left(\sum_{i=1}^n w_i x_i^2\right) - \left(\sum_{i=1}^n w_i x_i y_i\right) \left(\sum_{i=1}^n w_i x_i\right)}{\Delta}$$

$$\Delta = \left(\sum_{i=1}^n w_i\right) \left(\sum_{i=1}^n w_i x_i^2\right) - \left(\sum_{i=1}^n w_i x_i\right)^2$$

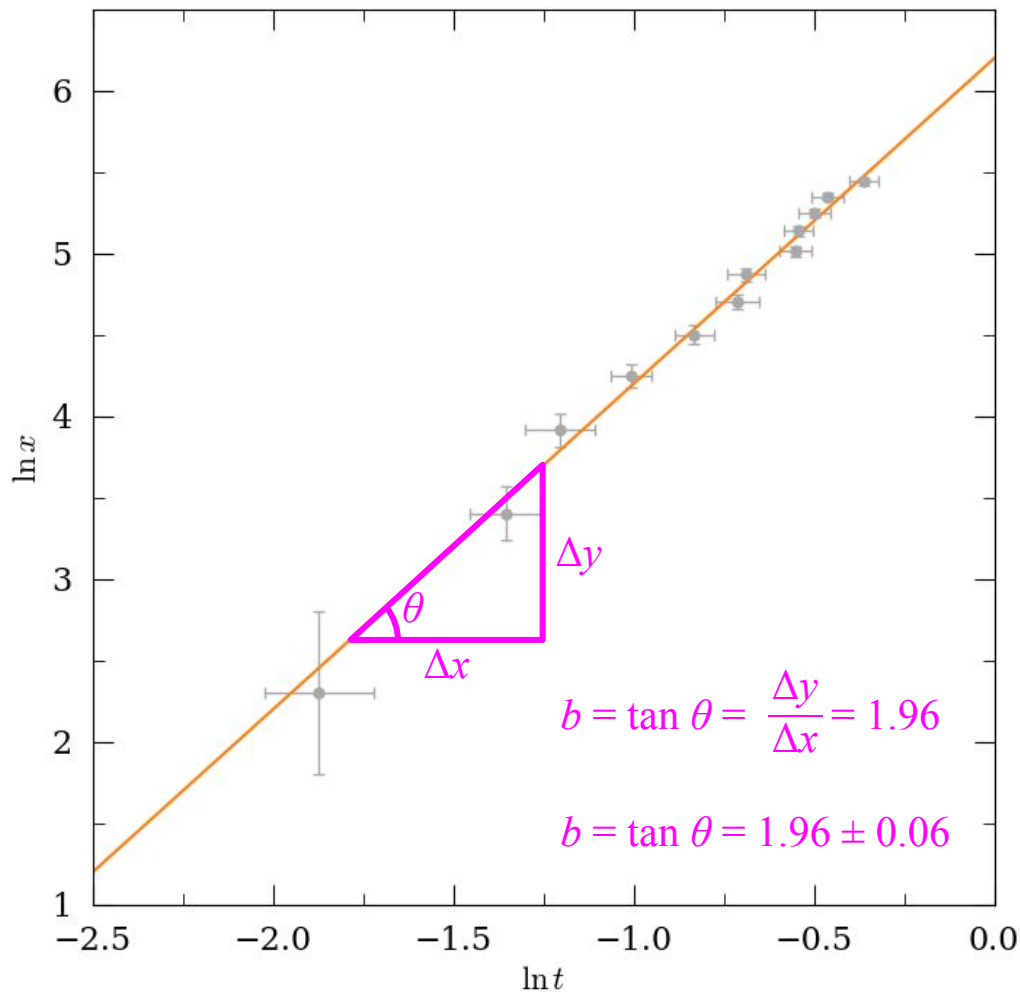
$$\sigma_a^2 = \frac{\sum_{i=1}^n w_i}{\Delta}$$

$$\sigma_b^2 = \frac{\sum_{i=1}^n w_i x_i^2}{\Delta}$$

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